

Response to Review: JCGS-25-420

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We thank the anonymous associate editor and reviewers for their careful reviews. What follows is our point-by-point response to reviewer comments. The original reviewer comment is in normal font and *our response is in italic text*.

Note: RMSE has been changed to HBE throughout the paper. See Reviewer 2 comments for the reason.

Reviewer 1

Summary:

This manuscript proposes an algorithm to evaluate nonlinear dimension reduction (NLDR) techniques. The authors identify some characteristics and limitations of NLDR and develop an algorithm to analyze NLDR results' quality of how well they match the original high-dimensional data.

Strengths:

1. This manuscript is well-organized, providing sufficient background knowledge and examples which makes this paper easy to follow.
2. This paper contains sufficient technical details in the introduction of the algorithm.

Weakness:

1. The main concern is whether the proposed algorithm can properly evaluate the NLDR's performance. Using the RMSE plot with different binwidth values (e.g., in Figure 8 and Figure 11) to evaluate NLDR's performance is problematic for not considering the density within each cluster. For NLDR results that have denser clusters, for example layout a,c,g,h in Figure 11, a smaller binwidth needs be used to achieve the same average count in each bin, which will cause their lines in Figure 11 to be more upper left. However, denser clusters are not worse NLDR results.

Thank you for this thoughtful comment. This could a problem when comparing the HBE curves for different layouts because the same binwidth could be correspond to very different average bin density. This is something that we have discussed at length in developing the methods, and decided that using binwidth as the horizontal variable is more faithful comparison than using average bin density, primarily because it is directly controlled by the number of bins. To make this clear we have added the HBE vs ave density plots to all figures where HBE vs a_1 is used. This shows the difficulty in using density to choose layouts, but it is helpful to see. Other diagnostic measures, proportion of non-empty bins and standardized bin count can also be useful in deciding the best fit. Ultimately, denser clusters in a layout occur when there are big separations between clusters. If this large separation is not present in the high dimensions the layout is not honest, and using binwidth more adequately reflects this.

2. In Figure 7, as RMSE steadily increases with the binwidth, it's not clear why the three binwidths 0.03, 0.05, and 0.07 are chosen. In addition, we shouldn't use a single simulated dataset to determine the default hyperparameters without any real analysis.

Thank you for raising this important point. The three binwidths (0.03, 0.05, and 0.07) represent small to moderate values, to illustrate exploring trade-offs between HBE, bin density and proportion of non-empty bins when decding which to use. This has been made clearer in the text.

The use of a single simulated dataset to illustrate a method is common practice. It needs to be sufficiently simple to communicate the method. The applications section has two examples that illustrate usage of the technique more generally. We have made this clearer in the text, when the simulated data is introduced.

3. For several figures that contains results from multiple NLDR algorithms, it would be nice to add the information of which algorithm/ what hyperparameter values for each of the subfigure. Many readers may have tried several NLDR algorithms themselves, and they will benefit from those figures to gain insights about behaviors of different NLDRs. Also it's nice to mention which data is used to generate each figure (e.g., in Figure 7).

Agreed. We have provided a table of the method and hyper-parameters used for every layout shown in the paper, in the appendix.

4. In section 3.6.1, it says “Values of b_1 between 2 and $b_1 = \sqrt{n/r_2}$ are allowed”, why the choice of b_1 should consider r_2 ?

The aspect ratio of the NLDR layout is typically preferred to be conserved when plotting, and hence when we bin the data. r_2 reflects this aspect ratio, because r_1 is fixed at 1, for ease of calculations. The range of recommended values for b_1 is adapted from the Diaconis-Freedman Rule, where the upper limit takes aspect ratio into account. It is just a recommendation. Practically, 2 is too few bins, and the upper bound is possibly too many. We have revised the text to make this clearer.

Reviewer 2

Summary

The manuscript proposes a novel, mostly graphical diagnostic to evaluate nonlinear dimension reduction (NLDR) layouts by effectively “lifting” the 2-D embedding back into the original high-dimensional space and inspecting the fit visually via tours (i.e., sequences of 2D-linear projections of the original high-D space). The goal is to help analysts discern which 2-D projection is the most “honest” (faithful) representation of the true structure in the data. The paper’s contributions are:

1. an algorithm that overlays a geometric model on an NLDR layout using hexagonal binning and Delaunay triangulation in the embedding, then maps this model into the original large-D data space for visual assessment;
2. a new goodness-of-fit metric, the RMSE in original space, computed as the root mean squared distance between each point and its bin centroid in high-D (Eq 2) (3) interactive visualization tools (implemented in published software) to compare/investigate different NLDRs

It is mostly well written, addresses an important problem with critical real-world relevance, and provides a very well done, extensively documented R implementation. However, this reviewer is skeptical about the suitability of the newly proposed GOF metric to choose the most “reasonable” or “faithful” NLDR, finds the empirical evaluation of it to be severely lacking without comparison to alternative measures from the literature, and would request some restructuring and shortening of the paper.

We have now included a comparison with existing metrics, which shows that it performs favorably, and is a suitable addition to the analyst’s toolbox.

Strengths

Relevance, Accessibility, and Intuitiveness

The way this method visualizes the spatial distortions induced by NLDR seems uniquely accessible to me, and, coupled with the modern interactive/animated visualizations implemented for it, seems likely to become a very useful tool to investigate and understand high-D data structures while avoiding over-reliance on / misinterpretation of misleading 2D patterns so prevalent in many areas at the moment (e.g. [Izarry \(2024\)](#), [Chari & Pachter \(2023\)](#)).

Thank you. We have added these references.

Background Material & Examples

The paper is fully fleshed out with illuminating examples, and I commend the authors for the effort they put into documenting their package and providing very informative videos etc of the method in action.

Reproducibility and Code Availability

The paper is exemplary in this regard - the underlying software is a CRAN compliant R package, and the Github repository itself allows to recompile the entire paper and appendix with all figures etc from scratch.

Major Issues

No Standard Embedding Quality Metrics or Comparisons

My primary concern is the absence of established quantitative metrics

1. to evaluate to what extent the newly proposed “RMSE” metric corresponds to established notion of embedding faithfulness and
2. to validate the different embeddings under consideration.

The paper’s only evaluation of “honesty” is done via the proposed custom “RMSE” in high dimensional space (defined in Section 3.4) and visual heuristics, but it does not report any conventional measures like trustworthiness/continuity scores, k -Neighborhood preservation rates (as derived from the co-ranking matrix - e.g. R_{NX} -curves for different neighborhood sizes k , c.f. ([Lee et al, 2015](#)), ([Lueks et al, 2011](#))), or Shepard diagrams of original space vs. embedding space distances. This is a significant omission because NLDR almost always faces a trade-off between optimizing preservation of local structures vs preservation of global

structures, which these metrics are (mostly) designed to resolve / analyze. For example, the authors could have computed such measures of trustworthiness for each NLDR layout to see if the one with lowest “RMSE” indeed had the highest (local and or global) neighborhood structure preservation but no such analysis was provided. The manuscript doesn’t even mention or discuss these metrics, which is not acceptable in a study focused on faithful representation, in my opinion.

This is valuable feedback. The revised manuscript has substantially changed to address this omission, and also includes material on local and global structure preservation.

Questions a revised version should aim to resolve:

1. does “RMSE” measure local or global structure preservation (or a mixture of both)? From the definition, it seems to be focused on how well small neighborhoods in embedding space (the bins) correspond to small neighborhoods in high-D space, with, IIUC, particularly high values if there are lots of “hard intrusions” (i.e., points that are embedded in the same 2D bin despite being far apart in high-D space)?

The HBE metric is intended to complement existing measures rather than replace them. When fine-grained bins are used, HBE primarily captures the preservation of local neighborhoods, reflecting how well small groups of points in the NLDR layout correspond to their high-dimensional counterparts. For coarser bins, the metric summarises more global structure. By examining the HBE vs binwidth for different layouts we can assess local and global fit.

2. how strongly are “RMSE” values correlated to other established measures of (local or global) structure preservation? and (how) are these correlations affected by the binning hyperparameters?

A comparison with existing metrics is included in the revision.

3. are the NLDRs selected by lowest “RMSE” the ones that would also be selected by other, more established measures of embedding faithfulness whose characteristics are well understood? at the very least, such measures should be provided, compared and discussed for all the embeddings under consideration in the case studies.

HBE is intended as a diagnostic tool rather than a definitive selection metric. It is an additional tool in the toolbox, but it is based on an ability to view NLDR model in the data space to examine how well high-dimensional structure is captured. A comparison with existing metrics is added, and shows how it can evaluate local and global structure preservation.

The reliance of “RMSE” on euclidean distances in high-D space also seems limiting:

1. at least for coarser bins and/or highly non-linear data structures, geodesic distances along the data manifold à la ISOMAP are likely to measure a more relevant kind of dissimilarity

2. a definition based on a more general notion of distance would make this applicable to investigating 2D embeddings of data types for which the euclidean distance of their numerical data representations is not a suitable/applicable measure of dissimilarity (e.g. images, text).

Agreed, but we think this is better left for future work. Distance along a manifold is a complicated and far from universally applicable quantity. For smaller binwidths, we are effectively computing Euclidean distance very locally which can capture highly nonlinear shapes. The problem, as you say, is for bigger binwidths, which may be better to ignore for many data sets.

Insufficient Validation of the Proposed Diagnostic (RMSE) Across Methods

Related to the above points - the paper has no theoretical or empirical demonstration that minimizing this “RMSE” aligns with minimizing other distortion measures like stress or maximizing neighborhood preservation (as measured by the co-ranking matrix). Furthermore, the claim that RMSE can be used to compare different NLDR methods on equal footing deserves scrutiny. Different methods optimize different criteria; it’s plausible that one method might (tends to) achieve lower “RMSE” simply by construction, especially if my reading that it mostly rewards embeddings that manage to avoid hard intrusions is correct. Overall, much more critical validation of the RMSE diagnostic is needed to convince readers that “honesty” as defined here aligns with meaningful goodness-of-fit in embedding.

Yes, point taken. The revision now includes a comparison with existing metrics, and illustrates for the examples shown that it is a reasonable addition. At some point, conducting a large simulation study with a range of specified structures would be useful. But the problem with tackling this at the moment is that there is no particular benchmark set of datasets for NLDR that have very well quantified structures. We are working towards this in other work. A package called cardinalR is a generator of benchmark shapes, that can be organised into more complex multi-cluster data sets. This could form the basis of a thorough simulation study on the strengths and weaknesses of different NLDR methods.

Limitations of the Evaluation (Simulation Study Design):

The manuscript includes a simulation to illustrate the approach (the “2NC7” dataset with two nonlinear clusters in 7-D) and two real-data applications (single-cell RNA-seq and an image dataset of handwritten digits). While these examples are appropriate, the scope of simulation scenarios is narrow, and the analysis lacks any notion of variability. The simulation seems to be a single realization of a specific two-cluster scenario. The authors did not explore, for instance, varying the overlap between clusters, adding more clusters or different manifold shapes, or introducing different noise levels. Ideally, the authors would systematically vary factors like: number of clusters or pattern types, intrinsic dimensionality of structures, sample size, etc., and show that their approach reliably selects the truth (or at least, the most interpretable view) in

each case, and how well alternative measures would do so. Because only one run is shown, there is also no information how stable the RMSE curves in Figures 8 and 13 are across replications. I encourage the authors to perform more systematic, replicated simulations, report measures of variability for their performance evaluations, compare against relevant alternatives, and take much more care not to overstate their conclusions beyond the tested scenarios (e.g. the claim that it “identify bad representations so they can be avoided” is based on a very limited set of examples).

This is not a simulation study. A single, simple yet sufficiently interesting simulated data set is used to bring the reader through the methodology. It is a standard approach for illustrating new methods. We have revised the text to make this clearer, and to explain why this particular pattern was used. At some point, agreed, running a benchmarking simulation study would be useful to do, but this should be future work, as explained above.

Minor Issues

1. First part of the title seems inappropriate for a scientific paper to me, as does the implied (and not substantiated, IMO) claim that the proposal is (primarily) useful to reliably select one of many NLDRs (as opposed to primarily useful for visually investigating details about a specific 2D NLDR).

The title has been modified, and we think it better reflects the content of the paper.

2. For marketing and clarity reasons, I would strongly suggest to not name the metric “RMSE”, but something more distinctive/descriptive.

Agreed. We changed to use “HBE” (hexbin error) instead of RMSE.

3. Flow from 3.1 to 3.2 (with too many sub-subsections) is difficult. Sections 3.2.1 (Scale the data) and 3.2.2 (Construct hexagon grid) are just short paragraphs, would be clearer if combined into a single numbered list of steps for the model construction. Much of 3 might be clearer and more compact if simply replaced by a direct step-by-step description of the algorithm in algorithmic pseudocode form with suitable short explanations.

We’ve kept this because with the other revisions done we think it still works to clearly delineate the steps in the algorithm.

4. Section 3.5 is a claim without supporting evidence – many out-of-sample embedding methods exist (some based on rather similar ideas, which should be referenced), you would need to show that this particular one actually works reasonably well for this to deserve space in the paper. I would strongly suggest to simply mention this in the discussion section as a possible avenue for further development instead.

We have added citations to related work, and explained this more carefully. This revision places the idea in a broader methodological context and highlights it as an avenue for future research.

5. Section 3.6 should be drastically shortened and details moved into an appendix.

We have actually expanded this section to address the feedback provided in this review. We think that this is important for understanding the new methods. We hope you agree with the careful new discussion about tuning the model fits to obtain useful layouts. To manage the length we have shifted the section on curiosities learned into the appendix.

6. p. 18: “A particular pattern that we commonly see is that analysts tend to pick layouts with clusters that have big separations between them.” The literature/internet contains many lively discussions of the perils of mis-interpreting and post-hoc’ing NLDRs (e.g. (Izarry (2024)), (Chari & Pachter (2023))), it might be good to cite/summarize these instead of/in addition to basing your motivation on personal anecdotal evidence? The tone here seems overly conversational to me (also: “we almost always see there are no big separations in the data”) and should be made more formal.

These references have been added, and the text is more carefully worded in the revision.

7. Section 4, esp. p.19: IMO, “to compare and assess a range of representations” an analyst really should look at their respective RNX curves, stress values, and Shepard diagrams, think about whether they care more about local or global structure preservation, and then decide based on those factors. The method proposed here seems to me to be mostly useful to then investigate specific regions / slices of high-D space to see for which sets of data points the high-D structure is (not) preserved well in the NLDR. I find this whole section to be vastly overselling the proposal – it ignores and implicitly discards most of the prior work in this field. Its intro paragraphs are also highly redundant with other content in the paper and should be cut.

This is done in the revised manuscript.

8. Section 5 should be moved into an appendix to shorten the paper somewhat. Figure in Section 5.1. could benefit from coloring the clusters in different colors so global structure preservation is also somewhat legible from the figures for the NLDRs (e.g. “is orange cluster between green and blue as in high-D or do they switch positions”). Section 5.2 might also have benefited from a corresponding ISOMAP or similar embedding of this data suitable for truthfully (isometrically!) unrolling the high-D manifold, and then showcasing that such a successful “unrolling” can be discerned from the way the projections look? tSNE just isn’t isometric, so the distortions visible here are entirely expected, IMO?

Section 5 has been shifted into the appendix.

Typos and grammar:

Revised version should be much more carefully proofread: “appropraite”, “doen’t”, “it is has two separated nonlinear clusters”, “the methods tNSE”

Thank you, and apologise for these errors. We have carefully spell-checked and proof-read the revised manuscript.